

it getting destroyed, or using a spray can to fill the area with smart motes? If some bots died, the others would rush back, warning of something evil going on down there. Need to destroy toxic waste? Let each swarmbot carry a little bit of it, and what you have is a toxic swarm...

The applications of a swarm of tiny, suitably-fitted robots are limited only by the imagination. But what's scary is: suppose that toxic swarm were to veer in the direction of Parliament House?

### Goo... D?

We're now back to grey goo, which, as you'll recollect, we started off with. The fear is about swarms getting out of control, clogging the atmosphere, and doing a variety of other bad things. The problem comes when we make the robots self-replicating. (There is good reason to do so, from the cost as well as redundancy points of view.) And once they can self-replicate, what would happen if something went wrong at the control centre? Would they multiply, swim through toxic waste, come into populated areas, and spread death and disease? Would they consume people?

The scare began when molecular nanotech pioneer Eric Drexler first used the term "grey goo" in his 1986 book *Engines of Creation*. In that book:

"Early assembler-based replicators could beat the most advanced modern organisms. 'Plants' with 'leaves' no more efficient than today's solar cells could out-compete real plants, crowding the biosphere with an inedible foliage. Tough, omnivorous 'bacteria' could spread like blowing pollen, replicate swiftly, and reduce the biosphere to dust in a matter of days."

Drexler doesn't now talk in such terms; he has recently made attempts to focus worry regarding nanotechnology in the right directions. But then some people read Michael Crichton's 2002 novel *Prey*, which talks about precisely the same thing—the threat of nanobots going beyond control, replicating like nobody's business, and generally creating a scene. Some of these readers got scared.

Bill Joy, co-founder of Sun Microsystems, once wrote a column in *Wired Magazine* titled *Why The Future Doesn't Need Us*. There, he famously stated that we're fools rushing in—into territory best left unexplored by the GNR technologies (genetics, nanotechnology, robotics). Joy, certainly no Luddite, gauging from his profile, goes so far as to say that there are some research areas we ought not to pursue because the consequences might be dire. If a man of Joy's stature is worried, what about the rest of us?

Grey goo seems more of a public scare than a real problem. But there do exist scientists trying to prove that grey goo is impossible, and pseudo-scientists who say the scenario is not entirely implausible.

Space does not permit a plausible-or-not discussion, but it's essentially another doomsday and big-brother scenario. As Pister puts it, "Yes, personal privacy is getting harder and harder to come by. Yes, you can hype Smart Dust as being great for big brother. Yawn. Every technology has a dark-side deal with it."

Yawn, yes. But Google "grey goo" anyway. It's interesting. ■

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